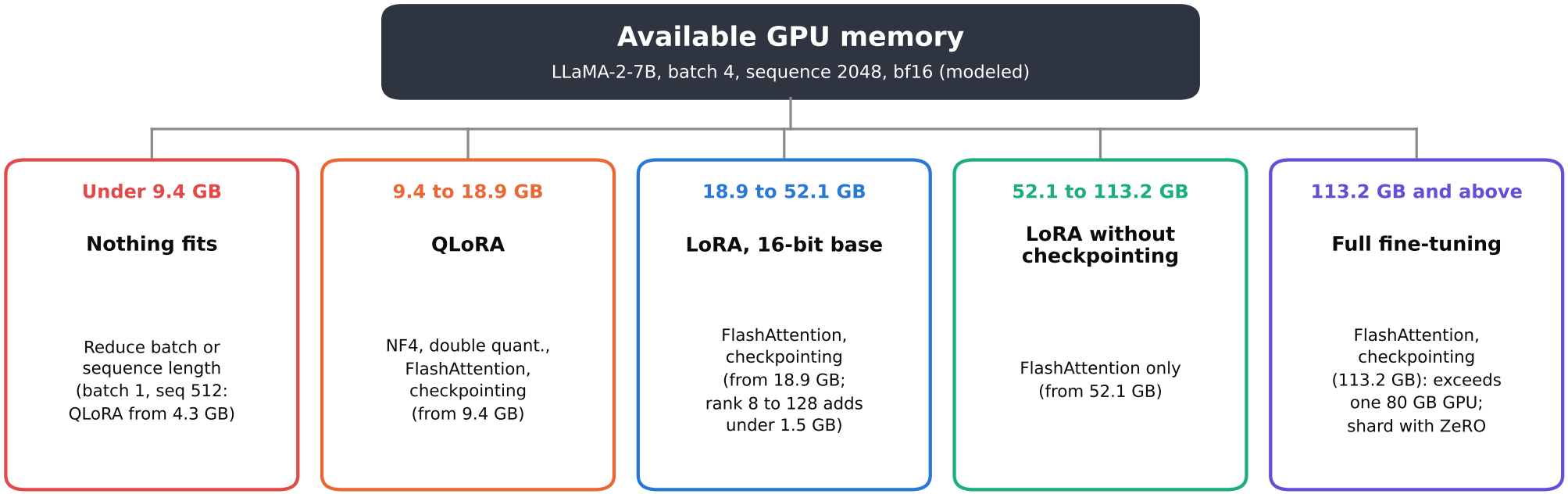


Starting hyperparameters reported in primary sources (not tuned recommendations)

- QLoRA: r = 64, alpha = 16, LoRA on all linear layers; LR 2e-4 (7B, 13B) or 1e-4 (33B, 65B); dropout 0.1 (up to 13B) or 0.05; max grad norm 0.3; constant LR schedule
  Dettmers et al. (2023), Appendix B.2 and Table 9; Guanaco instruction tuning, LLaMA 7B to 65B
- With LoRA on all linear layers, rank r had little effect on final performance
  Dettmers et al. (2023), Appendix A.1, Figure 4; LLaMA 7B, Alpaca
- On GPT-3 175B, ranks as small as 1 sufficed when adapting both Wq and Wv; Wq alone needed a larger r
  Hu et al. (2021), Table 6; WikiSQL and MultiNLI
- LoRA+: learning rate of B about 16x that of A for RoBERTa, about 2x to 4x for Llama-7B
  Hayou et al. (2024), Section 5.3; RoBERTa-base, GPT-2, Llama-7B
- In code and math domains LoRA substantially underperformed full fine-tuning in standard low-rank settings
  Biderman et al. (2024), abstract; Llama-2-7B, continued pre-training and instruction fine-tuning